Connected Bodies (Lifts)

Exam code: 9MA0

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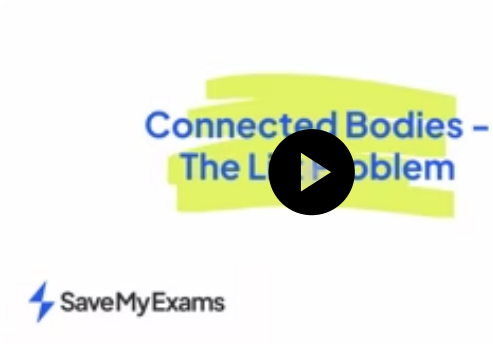

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Connected bodies (lifts)

Help me understand

What is a lift problem?

- A **lift problem** involves objects (particles) that are **directly** in **contact** with each other – typically a **person** or **crate** in a **lift**
- If it is not a person in the lift the object is often referred to as a **load**
- There may be **more** than **two objects** involved – for example **two crates** stacked on top of each other on a lift floor
- **Vertical** motion is involved so use g m s^{-2}, the **acceleration due** to **gravity**, where appropriate
  - Gravity always acts **vertically downwards**
  - Depending on the **positive** direction chosen – and which other forces are acting vertically – **acceleration** (a m s^{-2}) may be **positive** or **negative**
- Remember that **acceleration** links *F= ma* (**N2L**) and the '*suvat*' equations

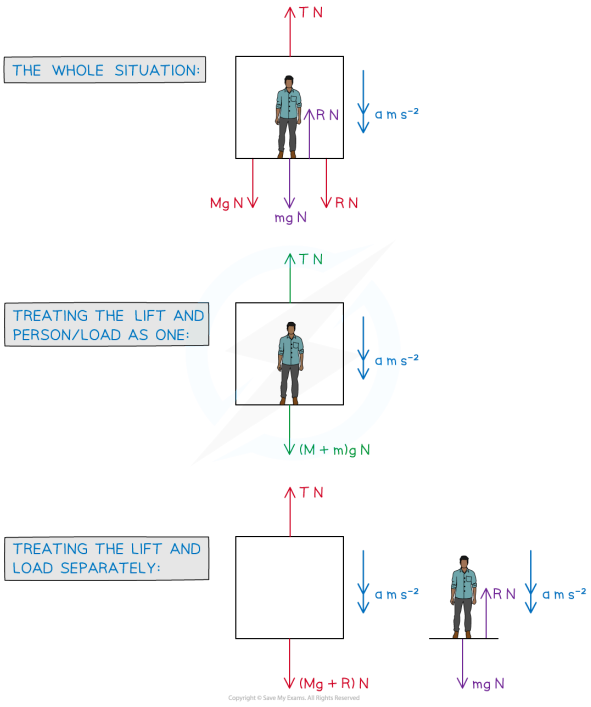
How do I solve 'lift problem' questions?

- Lift problems will only consider motion in the **vertical** direction
- As motion is involved **Newton's Laws of Motion** apply so use "*F = ma*" (N2L)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

direction the system can be **treated as one** particle (as well as separate particles)
    - There is no **reaction force** acting on the **lift** or **load** when treating the particle as one - mathematically they cancel each other out
    - You can think of the upward force as counteracting the person's weight **and** moving the load upwards; N3L applies so there must be an equal force acting in the opposite direction; - you can think of this as the force keeping the person in contact with the lift floor whilst it is moving

- For **constant acceleration** the '*suvat*' equations could be involved



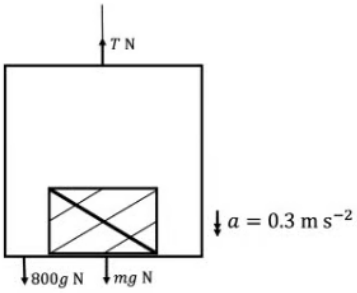
How do I find equations for problems involving lifts?

- Form the equations as follows:
    - Treating the lift and person/load as one
    - (↓) $(M + m)g - T = (M + m)a$
    - Treating the lift and person/load separately
      Lift: (↓) $(Mg + R) - T = Ma$
    - Person/load: (↓) $mg - R = ma$

- You do not necessarily need all equations but if in doubt attempt all and it may help you make progress

Worked Example

A lift of mass 800 kg is moving vertically downwards with acceleration 0.3 m s $^{-2}$. A load of mass *m* kg lies on the floor of

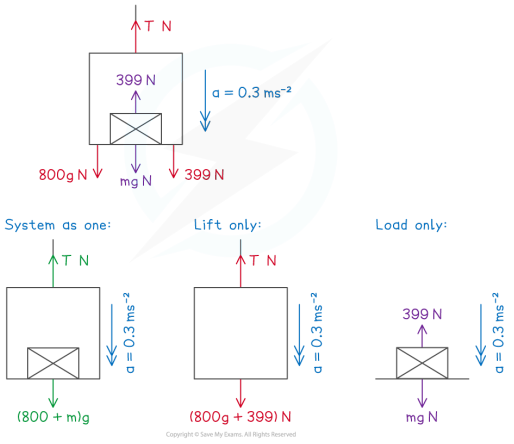


(a) Briefly explain how the force of 800g N arises in this problem.

Answer:

(a) The force 800g N is the weight of the lift. "W = mg" so the weight of the lift is the product of its mass, 800 kg, and g m s⁻², the acceleration due to gravity. Weight always acts vertically downwards.

Step 1: Sketch appropriate diagrams
(You can add to the given diagram)



(b) Find the mass of the load, m kg .

Answer:

(c) Find the tension, T N, in the cable of the lift.

Answer:

Examiner Tips and Tricks

- Sketch diagrams or add to any diagrams given in a question.
- If in doubt of how to start a problem, draw **all** diagrams and try writing an equation for each. This may help you make progress as well as picking up some marks.
- Watch out for "hidden lift" problems – we're not strictly talking elevators here! For example, a load being raised by a crane; the "lift" would be a platform (such as a pallet) and the "lift cable" would be the cable connecting the crane to the load. Another common alternative is a fast rising (or falling) fairground ride.

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